

AP CALCULUS AB – UNIT 1

Limits and Continuity

Expanded AP-Style Review: Full Coverage and Challenge



Name: _____ Class: _____ Date: _____

Directions

This review deliberately samples every website lesson and includes a larger proportion of difficult and challenge reasoning. Part A is non-calculator. Part B is calculator-permitted. Free-response work must show setup, exact reasoning, units, and theorem conditions.

Section I – Part A: No Calculator

1. Which statement is the precise meaning of $\lim_{x \rightarrow c} f(x) = L$?

- (A) There exists $\varepsilon > 0$ such that every $\delta > 0$ works (B) $f(c) = L$ and f is differentiable at c
 (C) For every $\delta > 0$ there exists $\varepsilon > 0$ depending on x (D) For every $\varepsilon > 0$ there exists $\delta > 0$ such that $0 < |x - c| < \delta$ implies $|f(x) - L| < \varepsilon$

2. To prove $\lim_{x \rightarrow 2} x^2 = 4$, which choice of δ is sufficient for every $\varepsilon > 0$?

- (A) 5ε (B) $\min\{5, 1/\varepsilon\}$
 (C) $\min\{1, \varepsilon/5\}$ (D) $\varepsilon + 5$

3. For $f(x) = x^2 - 4x$, which expression represents the instantaneous rate at $x = 3$?

- (A) $\frac{f(3) - f(3)}{3 - 3}$ (B) $\lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3}$
 (C) $\lim_{x \rightarrow 0} \frac{f(x) - f(3)}{x - 3}$ (D) $\frac{f(4) - f(2)}{2}$

4. A particle has $s(t) = 5t^2 - t$. Its instantaneous velocity at $t = 2$ is

- (A) 9 (B) 10
 (C) 19 (D) 20

5. For $P(t) = \frac{800}{1 + 3e^{-0.4t}}$, a calculator gives $P(4.99) \approx 568.330$ and $P(5.01) \approx 569.645$. Which is the best estimate of $P'(5)$?

- (A) 0.658 (B) 6.58
 (C) 65.8 (D) 131.5

6. If $\lim_{x \rightarrow 2^-} f(x) = 5$ and $\lim_{x \rightarrow 2^+} f(x) = 5$, which must be true?

- (A) $f(2) = 5$ (B) $f(2)$ exists
 (C) $\lim_{x \rightarrow 2} f(x) = 5$ (D) f is differentiable at 2

7. Which notation means that $g(x)$ decreases without bound as x approaches 4 through values less than 4?

- (A) $\lim_{x \rightarrow 4^-} g(x) = -\infty$ (B) $\lim_{x \rightarrow 4^+} g(x) = -\infty$
 (C) $\lim_{x \rightarrow -4^-} g(x) = +\infty$ (D) $g(4) = -\infty$

8. A calculator table produces the data below. Which notation best describes the behavior?

x	1.99	1.999	2.001	2.01
$g(x)$	-48.7	-498.7	501.3	51.3

- (A) $\lim_{x \rightarrow 2} g(x) = 0$ (B) $\lim_{x \rightarrow 2^-} g(x) = -\infty$ and $\lim_{x \rightarrow 2^+} g(x) = +\infty$
 (C) $\lim_{x \rightarrow 2} g(x) = +\infty$ (D) $g(2) = 0$

9. If $\lim f = 2$ and $\lim g = -5$, then $\lim(3f - g^2)$ equals

- (A) -19 (B) -7
(C) 19 (D) 31

10. Which condition is required to use the quotient law for $\lim f(x)/g(x)$?

- (A) $f(c)$ exists (B) $g(c) \neq 0$
(C) $\lim g(x) \neq 0$ (D) f and g are differentiable

11. A calculator gives $u(x) \rightarrow 1.73205$ and $v(x) \rightarrow -0.50000$. Estimate $\lim[u(x)^2 + 4v(x)]$.

- (A) -1 (B) 0
(C) 1 (D) 5

12. $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$ equals

- (A) 0 (B) 3
(C) 6 (D) DNE

13. $\lim_{x \rightarrow 0} \frac{\sin(4x)}{3x}$ equals

- (A) $\frac{3}{4}$ (B) $\frac{4}{3}$
(C) 3 (D) 4

14. A calculator suggests $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{x} \approx 2$. Which table best supports the conclusion?

- (A) Only $x = 0$ (B) Inputs near 0 from both sides with increasing precision
(C) Only positive integers (D) Only $x = 1, 2, 3$

15. After substitution, $\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$ has form $7/0$. The best next step is to

- (A) cancel 7 (B) inspect denominator signs from each side
(C) report 0 (D) use the product law

16. Which procedure is most appropriate for $\lim_{x \rightarrow 9} \frac{x-9}{\sqrt{x}-3}$?

- (A) Long division (B) Conjugate rationalization
(C) Squeeze theorem (D) Numerical integration

17. A calculator graph and table disagree near $x = 2$ because the graph window is coarse. Which evidence should be refined first?

- (A) The title of the function (B) The local graph window and table precision
(C) The function value at $x = 0$ (D) A distant interval

18. If $-2|x - 1| \leq f(x) \leq 5|x - 1|$ near 1, then $\lim_{x \rightarrow 1} f(x)$ is

- (A) -2 (B) 0
(C) 5 (D) not determined

19. Which bound is useful for $x^3 \cos(1/x)$ near 0?

- (A) $|x^3 \cos(1/x)| \leq |x|^3$ (B) $|x^3 \cos(1/x)| \geq 1$
(C) $\cos(1/x) \rightarrow 0$ (D) $x^3 \geq 1$

20. A graphing calculator shows violent oscillation for $x \sin(1/x)$ near 0, but the envelope narrows to the axis. The limit is

- (A) -1 (B) 0
(C) 1 (D) DNE

21. Which evidence identifies a jump discontinuity at $x = c$?

- (A) Both sides approach the same finite value.
- (B) Finite one-sided limits exist but are unequal.
- (C) Both sides approach $+\infty$.
- (D) The function value is missing.

22. For $f(x) = \frac{(x-2)^2}{(x-2)^3}$, the discontinuity at 2 is

- (A) removable
- (B) jump
- (C) infinite
- (D) none

23. A calculator graph of $\frac{x^2-4}{x^2-x-2}$ seems smooth except near $x = -1$ and $x = 2$. The correct classification is

- (A) holes at both points
- (B) poles at both points
- (C) a pole at -1 and a hole at 2
- (D) a hole at -1 and a pole at 2

24. Which condition is not required for continuity at c ?

- (A) $f(c)$ exists
- (B) $\lim_{x \rightarrow c} f(x)$ exists
- (C) $\lim_{x \rightarrow c} f(x) = f(c)$
- (D) $f'(c)$ exists

Section I – Part B: Calculator Permitted

- 25.** At $x = a$, a graph approaches 6 from both sides but has a filled point at height -3 . Which statement is true?
- (A) $\lim_{x \rightarrow a} f(x) = -3$ (B) $\lim_{x \rightarrow a} f(x) = 6$
 (C) The limit DNE. (D) $f(a) = 6$
- 26.** Which graphical feature necessarily prevents a finite two-sided limit at the target?
- (A) A corner (B) A cusp whose sides meet
 (C) A jump with unequal one-sided heights (D) A hole
- 27.** A calculator graph suggests a limit of 1.732 at $x = 2$, but the curve is steep and the window uses y-scale 10. What is the best next step?
- (A) Report 1.732000 immediately. (B) Zoom and confirm with a numerical table from both sides.
 (C) Read the filled point only. (D) Conclude DNE.
- 28.** A table near $x = 5$ has outputs approaching -2 from both sides, while the center row is 17. Which conclusion is best?
- (A) $\lim_{x \rightarrow 5} f(x) = 17$ (B) $\lim_{x \rightarrow 5} f(x) = -2$
 (C) The limit DNE. (D) No conclusion is possible.
- 29.** Which input list is best for estimating $\lim_{x \rightarrow 3} f(x)$?
- (A) 0, 1, 2, 3 (B) 2.9, 2.99, 3.01, 3.1
 (C) 3, 4, 5, 6 (D) $-3, -2, -1, 0$
- 30.** A calculator table for $h(x) = \frac{e^x - 1}{x}$ gives 0.99950, 0.99995, 1.00005, 1.00050 near $x = 0$. The best estimate is
- (A) 0 (B) 0.5
 (C) 1 (D) DNE
- 31.** Which pair of statements is coherent?
- (A) $\lim_{x \rightarrow 2} f(x) = 3$ and a graph approaching 5 from both sides (B) $f(2) = 7$ and a filled point at $(2, 7)$
 (C) $\lim_{x \rightarrow 2^-} f(x) = 1$ and a left table approaching 4 (D) A formula equal to $x + 1$ near 2 and a hole at height 8
- 32.** A table approaches 6 from both sides while the formula is undefined at the target. Which is the best conclusion?
- (A) The limit must be undefined. (B) The limit is estimated as 6 and the point value is undefined.
 (C) The point value is 6. (D) The limit is infinite.
- 33.** A graphing calculator displays a line through $x = 3$ for $f(x) = \frac{x^2 - 9}{x - 3}$. What information may be hidden?
- (A) A removable hole at $x = 3$ (B) A vertical asymptote at $x = 0$
 (C) A jump at every integer (D) A horizontal asymptote at $y = 0$
- 34.** $\lim_{x \rightarrow \infty} \frac{4x^2 + 1}{2x^2 - 3x}$ equals
- (A) 0 (B) 2
 (C) 4 (D) $+\infty$
- 35.** $\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 + 1}}{x}$ equals
- (A) -1 (B) 0
 (C) 1 (D) $+\infty$

36. A calculator table for $f(x) = \frac{2x^3+1}{x^3-4}$ at large positive and negative inputs approaches

(A) -2

(B) 0

(C) 2

(D) $+\infty$

Section II – Free Response

Questions 1–2 may be completed with a calculator. Questions 3–4 are non-calculator.

FRQ1 – Limits from numerical and graphical evidence

A function is sampled near $x = 2$ with

x	1.9	1.99	1.999	2.001	2.01	2.1
$f(x)$	3.61	3.9601	3.996001	4.004001	4.0401	4.41

(a) Estimate $\lim_{x \rightarrow 2} f(x)$ and justify from both sides. [2 points]

(b) Give a simple formula consistent with the table and state whether $f(2)$ is determined by the data. [2 points]

(c) Explain one way rounding could create a misleading conclusion. [2 points]

FRQ2 – Algebraic limit and continuity repair

Let $g(x) = \frac{x^2 - 5x + 6}{x - 2}$ for $x \neq 2$.

(a) Evaluate $\lim_{x \rightarrow 2} g(x)$ and identify the algebraic procedure. [3 points]

(b) Define $g(2)$ to make the extended function continuous. [1 points]

(c) Explain why cancellation does not prove the original function was defined at 2. [2 points]

FRQ3 – Precise definition of a limit

Prove from the ε - δ definition that $\lim_{x \rightarrow 2} x^2 = 4$.

(a) Explain why imposing $|x - 2| < 1$ produces a useful bound on $|x + 2|$. [2 points]

(b) Choose an explicit $\delta(\varepsilon)$ and complete the proof. [4 points]

(c) State why the condition $0 < |x - 2|$ is present. [1 points]

FRQ4 – Squeeze theorem with shrinking oscillation

Consider $h(x) = x^2 \sin(1/x^3) \cos(e^{1/x^4})$ for $x \neq 0$.

(a) Find a two-sided bound for $|h(x)|$. [2 points]

(b) Evaluate $\lim_{x \rightarrow 0} h(x)$ with a complete theorem justification. [3 points]

(c) Explain why the oscillatory factors alone do not have limits but the product does. [2 points]